

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Develop a Dynamic Stranger Things Event Registration Webpage using JavaScript and DOM

2. AIM

To develop a dynamic event registration webpage using HTML, CSS and JavaScript DOM manipulation that validates user input and displays the submitted registration details dynamically.

3. OBJECTIVE

The objectives of this experiment are:

- Create a styled Stranger Things event registration interface.
 - Use HTML input and select elements to collect registration details.
 - Use JavaScript functions to handle registration and reset actions.
 - Access and modify HTML elements using `document.getElementById()`.
 - Validate that all required fields contain values before registration.
 - Generate a random registration ID using `Math.random()` and `Math.floor()`.
 - Display the entered name, email, team, event and current date/time dynamically.
 - Clear the form and hide the result section using the reset operation.
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4. THEORY

JavaScript is a scripting language used to add dynamic behaviour and interactivity to web pages. In this program, JavaScript responds to button clicks, reads values from form controls, validates the input and updates the result section.

Document Object Model (DOM) represents the HTML page as objects that JavaScript can access and modify. The program uses `document.getElementById()` to retrieve elements by their IDs and changes their values, display state and `innerHTML`.

JavaScript features used in the program include `let` variables, functions, conditional validation, `Math.random()`, `Math.floor()`, the Date object, `toLocaleString()`, string concatenation and dynamic HTML construction.

CSS features used in the interface include a dark gradient background, flexbox alignment, a bordered container, red glow effects, input and select styling, hover and focus states, and a fade animation.

5. ALGORITHM / PROCEDURE

1. Create the HTML document and define the viewport metadata and page title.
 2. Design the Stranger Things registration interface using headings, input fields, select elements and buttons.
 3. Apply CSS styling to the page, form controls, buttons and result section.
 4. Define the `register()` function and obtain name, email, team and event values using `document.getElementById()`.
 5. Trim the name and email values and check whether any required field is empty.
 6. If a required field is empty, display an alert asking the user to fill all fields and stop the registration operation.
 7. Generate a registration ID using a random four-digit number with the ST- prefix.
 8. Create a Date object and obtain the local date and time.
 9. Make the result element visible and insert the registration details using `innerHTML`.
 10. Display the registration-success message and welcome message.
 11. Define `resetForm()` to clear the input values, reset the selections and hide the result.
 12. Open the page in a web browser and verify the registration and reset behaviour.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Stranger Things Event Registration</title>
```

```
<style>
```

```
*{
    margin:0;    padding:0;
box-sizing:border-box;
font-family:Arial, sans-serif;

}
```

```
body{    background:linear-gradient(to
bottom,#000,#1a0000,#000);    color:white;    display:flex;
justify-content:center;    align-items:center;
min-height:100vh;
```

```
}
```

```
.container{    width:450px;
background:rgba(20,20,20,0.95);
padding:25px;    border-radius:15px;
```

```
    box-shadow:0 0 20px red;
border:2px solid crimson;
}
```

```
h1{
    text-align:center;
color:#ff3333;
text-shadow:0 0 10px red;
margin-bottom:5px; }
```

```
p{
    text-align:center;
color:#ddd;
margin-bottom:20px;
}
input,select{
```

```
        width:100%;
padding:12px;
margin:10px 0;
border:none;
border-radius:6px;
background:#2d2d2d;
color:white;
font-size:15px;
}

input:focus,
select:focus{
outline:none;
    box-shadow:0 0 10px red;
}

button{    width:48%;
padding:12px;
border:none;
border-radius:6px;
margin-top:15px;
cursor:pointer;
font-size:15px;

}

.register{
background:crimson;
color:white;
}

.register:hover{
background:red;
}

.reset{
background:#444;
color:white;
}

.reset:hover{
background:#666;
```

```
}
```

```
#result{      display:none;
margin-top:20px;
padding:15px;
border-radius:10px;
background:#111;
border:1px solid red;
animation:fade 0.5s;
```

```
}
```

```
@keyframes fade{ from
{opacity:0;transform:scale(0.9);} to
{opacity:1;transform:scale(1);}
}
```

```
.success{
color:#00ff99;
font-size:20px;
text-align:center;
}
</style>
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h1>STRANGER THINGS</h1>
```

```
<p>Upside Down Event Registration</p>
```

```
<input type="text" id="name" placeholder="Your Name">
```

```
<input type="email" id="email" placeholder="Email">
```

```
<select id="house">
```

```
<option value="">Choose Your Team</option>
```

```
<option>Hawkins Heroes</option>
```

```
<option>Hellfire Club</option>
```

```
<option>Scoops Troop</option>
```

```
<option>Upside Down Explorers</option>
```

```
</select>
```

```

<select id="event">
<option value="">Choose Event</option>
<option>Escape the Upside Down</option>
<option>Demogorgon Hunt</option>
<option>80's Costume Contest</option>
<option>Trivia Challenge</option>
</select>

<div style="display:flex;justify-content:space-between;">
<button class="register" onclick="register()">Register</button>
<button class="reset" onclick="resetForm()">Reset</button>
</div>

<div id="result"></div>

</div>

<script>

function register(){

let name=document.getElementById("name").value.trim(); let
email=document.getElementById("email").value.trim(); let
team=document.getElementById("house").value; let
event=document.getElementById("event").value; if(name=="
|| email==" || team==" || event==""){
    alert("Please fill all fields.");
return;
}

let id="ST-"+Math.floor(1000+Math.random()*9000); let

now=new Date();

document.getElementById("result").style.display="block";
document.getElementById("result").innerHTML=
"<h2 class='success'> Registration Successful!</h2><br>"+
"<b>Registration ID:</b> "+id+"<br><br>"+
"<b>Name:</b> "+name+"<br>"+
"<b>Email:</b> "+email+"<br>"+
"<b>Team:</b> "+team+"<br>"+
"<b>Event:</b> "+event+"<br>"+

```

```
"<b>Date:</b> "+now.toLocaleString()+"<br><br>"+
"<span style='color:#ff5555;'>Welcome, "+name+
"! The gate to the <b>Upside Down</b> awaits you...</span>";
```

```
}
```

```
function resetForm(){ document.getElementById
("name").value=""; document.getElementById
("email").value=""; document.getElementById
("house").selectedIndex=0; document.getElementById
("event").selectedIndex=0; document.getElementById
("result").style.display="none";
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

7. CODE EXPLANATION

Code	Explanation
HTML document structure	Defines the page with the DOCTYPE, html, head and body elements, along with meta tags and the page title.
CSS styling	Uses selectors, a dark linear gradient, flexbox, colours, borders, shadows, focus and hover states, and a fade animation.
Form controls	Uses text and email input fields and two select elements for choosing a team and event.
JavaScript functions	Defines register() for registration processing and resetForm() for clearing and resetting the form.
DOM manipulation	Uses document.getElementById() to access the form controls and the result element.
Input validation	Checks whether name, email, team or event is empty. If any field is empty, an alert is displayed and the function returns.
Random registration ID	Uses Math.random() and Math.floor() to generate a four-digit number with the ST- prefix.
Date and time	Creates a Date object and uses toLocaleString() to display the current local date and time.
Dynamic result	Changes the result element to visible and uses innerHTML to insert the registration ID and submitted details.
Reset operation	Clears the input values, resets both select elements to their first option and hides the result element.

8. TEST CASES AND EXECUTION DATA

The following test cases are based directly on the validation, registration and reset logic present in the source program.

Test Case	Input / Action	Expected Result	Status
TC-01	Leave one or more fields empty and click Register.	Alert "Please fill all fields." is displayed and registration is stopped.	Pass
TC-02	Enter name, email, team and event and click Register.	Registration result is displayed with generated ID, entered details and current date/time.	Pass

Test Case	Input / Action	Expected Result	Status
TC-03	After registration, click Reset.	Input fields are cleared, selections are reset and the result is hidden.	Pass
TC-04	Change the selected team and event and register with valid inputs.	The selected team and event appear in the dynamic result.	Pass

9. OUTPUT

Output 1: Stranger Things Event Registration after successful registration

The screenshot displays a web form titled "STRANGER THINGS" with the subtitle "Upside Down Event Registration". The form contains four input fields with the following values: "vijaya sri", "vijayadadhirao@gmail.com", "Scoops Troop", and "80's Costume Contest". Below these fields are two buttons: a red "Register" button and a grey "Reset" button. A green success message "Registration Successful!" is shown, followed by the registration details: "Registration ID: ST-4987", "Name: vijaya sri", "Email: vijayadadhirao@gmail.com", "Team: Scoops Troop", "Event: 80's Costume Contest", and "Date: 8/23/2026, 5:40:23 PM". A red welcome message at the bottom reads: "Welcome, vijaya sri! The gate to the Upside Down awaits you..."

The output shows the completed registration form with the entered details, selected team and event, generated registration ID, current date/time and dynamic welcome message.

10. RESULT

Thus, a dynamic Stranger Things Event Registration webpage was successfully developed using HTML, CSS and JavaScript. The program validates required inputs, generates a registration ID, displays submitted details dynamically using DOM manipulation, and provides a reset operation.